

Tutorial on Boolean Satisfiability (SAT) Solving

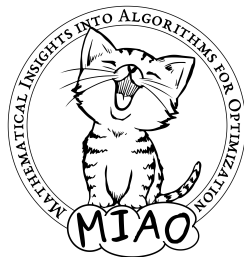
Jakob Nordström

University of Copenhagen and Lund University

*1st International Workshop on
Solving Linear Optimization Problems
for Pseudo-Booleans and Yonder*

Lund, Sweden

November 5, 2024



This Is Me...

Jakob Nordström

Professor

University of Copenhagen
and Lund University

www.jakobnordstrom.se



... And This Is What I Do for a Living

$$\begin{aligned} & (x_{1,1} \vee x_{1,2} \vee x_{1,3} \vee x_{1,4} \vee x_{1,5} \vee x_{1,6} \vee x_{1,7}) \wedge (x_{2,1} \vee x_{2,2} \vee x_{2,3} \vee x_{2,4} \vee x_{2,5} \vee x_{2,6} \vee x_{2,7}) \wedge (x_{3,1} \vee x_{3,2} \vee x_{3,3} \vee x_{3,4} \vee \\ & x_{3,5} \vee x_{3,6} \vee x_{3,7}) \wedge (x_{4,1} \vee x_{4,2} \vee x_{4,3} \vee x_{4,4} \vee x_{4,5} \vee x_{4,6} \vee x_{4,7}) \wedge (x_{5,1} \vee x_{5,2} \vee x_{5,3} \vee x_{5,4} \vee x_{5,5} \vee x_{5,6} \vee x_{5,7}) \wedge (x_{6,1} \vee \\ & x_{6,2} \vee x_{6,3} \vee x_{6,4} \vee x_{6,5} \vee x_{6,6} \vee x_{6,7}) \wedge (x_{7,1} \vee x_{7,2} \vee x_{7,3} \vee x_{7,4} \vee x_{7,5} \vee x_{7,6} \vee x_{7,7}) \wedge (x_{8,1} \vee x_{8,2} \vee x_{8,3} \vee x_{8,4} \vee x_{8,5} \vee x_{8,6} \vee \\ & x_{8,7}) \wedge (\neg x_{1,1} \vee \neg x_{2,1}) \wedge (\neg x_{1,1} \vee \neg x_{3,1}) \wedge (\neg x_{1,1} \vee \neg x_{4,1}) \wedge (\neg x_{1,1} \vee \neg x_{5,1}) \wedge (\neg x_{1,1} \vee \neg x_{6,1}) \wedge (\neg x_{1,1} \vee \neg x_{7,1}) \wedge (\neg x_{1,1} \vee \\ & \neg x_{8,1}) \wedge (\neg x_{2,1} \vee \neg x_{3,1}) \wedge (\neg x_{2,1} \vee \neg x_{4,1}) \wedge (\neg x_{2,1} \vee \neg x_{5,1}) \wedge (\neg x_{2,1} \vee \neg x_{6,1}) \wedge (\neg x_{2,1} \vee \neg x_{7,1}) \wedge (\neg x_{2,1} \vee \neg x_{8,1}) \wedge \\ & (\neg x_{3,1} \vee \neg x_{4,1}) \wedge (\neg x_{3,1} \vee \neg x_{5,1}) \wedge (\neg x_{3,1} \vee \neg x_{6,1}) \wedge (\neg x_{3,1} \vee \neg x_{7,1}) \wedge (\neg x_{3,1} \vee \neg x_{8,1}) \wedge (\neg x_{4,1} \vee \neg x_{5,1}) \wedge (\neg x_{4,1} \vee \\ & \neg x_{6,1}) \wedge (\neg x_{4,1} \vee \neg x_{7,1}) \wedge (\neg x_{4,1} \vee \neg x_{8,1}) \wedge (\neg x_{5,1} \vee \neg x_{6,1}) \wedge (\neg x_{5,1} \vee \neg x_{7,1}) \wedge (\neg x_{5,1} \vee \neg x_{8,1}) \wedge (\neg x_{6,1} \vee \neg x_{7,1}) \wedge \\ & (\neg x_{6,1} \vee \neg x_{8,1}) \wedge (\neg x_{7,1} \vee \neg x_{8,1}) \wedge (\neg x_{1,2} \vee \neg x_{2,2}) \wedge (\neg x_{1,2} \vee \neg x_{3,2}) \wedge (\neg x_{1,2} \vee \neg x_{4,2}) \wedge (\neg x_{1,2} \vee \neg x_{5,2}) \wedge (\neg x_{1,2} \vee \neg x_{6,2}) \wedge \\ & (\neg x_{1,2} \vee \neg x_{7,2}) \wedge (\neg x_{1,2} \vee \neg x_{8,2}) \wedge (\neg x_{2,2} \vee \neg x_{3,2}) \wedge (\neg x_{2,2} \vee \neg x_{4,2}) \wedge (\neg x_{2,2} \vee \neg x_{5,2}) \wedge (\neg x_{2,2} \vee \neg x_{6,2}) \wedge (\neg x_{2,2} \vee \neg x_{7,2}) \wedge \\ & (\neg x_{2,2} \vee \neg x_{8,2}) \wedge (\neg x_{3,2} \vee \neg x_{4,2}) \wedge (\neg x_{3,2} \vee \neg x_{5,2}) \wedge (\neg x_{3,2} \vee \neg x_{6,2}) \wedge (\neg x_{3,2} \vee \neg x_{7,2}) \wedge (\neg x_{3,2} \vee \neg x_{8,2}) \wedge (\neg x_{4,2} \vee \neg x_{5,2}) \wedge \\ & (\neg x_{4,2} \vee \neg x_{6,2}) \wedge (\neg x_{4,2} \vee \neg x_{7,2}) \wedge (\neg x_{4,2} \vee \neg x_{8,2}) \wedge (\neg x_{5,2} \vee \neg x_{6,2}) \wedge (\neg x_{5,2} \vee \neg x_{7,2}) \wedge (\neg x_{5,2} \vee \neg x_{8,2}) \wedge (\neg x_{6,2} \vee \neg x_{7,2}) \wedge \\ & (\neg x_{6,2} \vee \neg x_{8,2}) \wedge (\neg x_{7,2} \vee \neg x_{8,2}) \wedge (\neg x_{1,3} \vee \neg x_{2,3}) \wedge (\neg x_{1,3} \vee \neg x_{3,3}) \wedge (\neg x_{1,3} \vee \neg x_{4,3}) \wedge (\neg x_{1,3} \vee \neg x_{5,3}) \wedge (\neg x_{1,3} \vee \neg x_{6,3}) \wedge \\ & (\neg x_{1,3} \vee \neg x_{7,3}) \wedge (\neg x_{1,3} \vee \neg x_{8,3}) \wedge (\neg x_{2,3} \vee \neg x_{3,3}) \wedge (\neg x_{2,3} \vee \neg x_{4,3}) \wedge (\neg x_{2,3} \vee \neg x_{5,3}) \wedge (\neg x_{2,3} \vee \neg x_{6,3}) \wedge (\neg x_{2,3} \vee \neg x_{7,3}) \wedge \\ & (\neg x_{2,3} \vee \neg x_{8,3}) \wedge (\neg x_{3,3} \vee \neg x_{4,3}) \wedge (\neg x_{3,3} \vee \neg x_{5,3}) \wedge (\neg x_{3,3} \vee \neg x_{6,3}) \wedge (\neg x_{3,3} \vee \neg x_{7,3}) \wedge (\neg x_{3,3} \vee \neg x_{8,3}) \wedge (\neg x_{4,3} \vee \\ & \neg x_{5,3}) \wedge (\neg x_{4,3} \vee \neg x_{6,3}) \wedge (\neg x_{4,3} \vee \neg x_{7,3}) \wedge (\neg x_{4,3} \vee \neg x_{8,3}) \wedge (\neg x_{5,3} \vee \neg x_{6,3}) \wedge (\neg x_{5,3} \vee \neg x_{7,3}) \wedge (\neg x_{5,3} \vee \neg x_{8,3}) \wedge \\ & (\neg x_{6,3} \vee \neg x_{7,3}) \wedge (\neg x_{6,3} \vee \neg x_{8,3}) \wedge (\neg x_{7,3} \vee \neg x_{8,3}) \end{aligned}$$

Three Simple Problems. . .

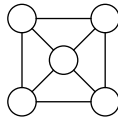
COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

Three Simple Problems...

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

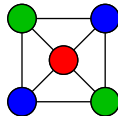


3-colouring?

Three Simple Problems. . .

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

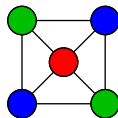


3-colouring? Yes

Three Simple Problems...

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?



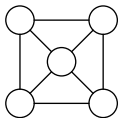
3-colouring? Yes, but no 2-colouring

Three Simple Problems. . .

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

Three Simple Problems...

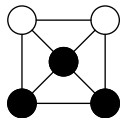


3-clique?

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

Three Simple Problems...

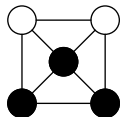


3-clique? Yes

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

Three Simple Problems...



3-clique? Yes, but no 4-clique

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

Three Simple Problems...

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

SAT

Given propositional logic formula, is there a **satisfying assignment**?

Three Simple Problems...

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

SAT

Given propositional logic formula, is there a **satisfying assignment**?

$$(x \vee z) \wedge (y \vee \neg z) \wedge (x \vee \neg y \vee u) \wedge (\neg y \vee \neg u) \\ \wedge (u \vee v) \wedge (\neg x \vee \neg v) \wedge (\neg u \vee w) \wedge (\neg x \vee \neg u \vee \neg w)$$

Three Simple Problems...

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

SAT

Given propositional logic formula, is there a **satisfying assignment**?

$$(x \vee z) \wedge (y \vee \neg z) \wedge (x \vee \neg y \vee u) \wedge (\neg y \vee \neg u) \\ \wedge (u \vee v) \wedge (\neg x \vee \neg v) \wedge (\neg u \vee w) \wedge (\neg x \vee \neg u \vee \neg w)$$

- Variables should be set to **true** or **false**
- Constraint $(x \vee \neg y \vee z)$: means x or z should be true or y false
- $\wedge$ means all constraints should hold simultaneously
- Is there a truth value assignment satisfying all constraints?

Three Simple Problems...

COLOURING

Does the graph $G = (V, E)$ have a **colouring** with k colours such that all neighbours have distinct colours?

CLIQUE

Is there a **clique** in the graph $G = (V, E)$ with k vertices that are all pairwise connected by edges in E ?

SAT

Given propositional logic formula, is there a **satisfying assignment**?

COLOURING: frequency allocation for mobile base stations

CLIQUE: bioinformatics, computational chemistry

SAT: easily models these and many other problems

...with Huge Practical Implications

- Some more examples of problems that can be encoded as propositional logic formulas:
 - computer hardware verification
 - computer software testing
 - artificial intelligence
 - cryptography
 - bioinformatics
 - et cetera...
- Leads to **humongous formulas** (100,000s or even 1,000,000s of variables)
- Can we use computers to solve these problems efficiently?
- Question mentioned already in Gödel's famous letter in 1956 to von Neumann (the "father of computer science")
- Topic of intense research in computer science ever since 1960s

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems
- But... There are also small formulas (just ~ 100 variables) that are completely beyond reach of even the very best SAT solvers

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems
- But... There are also small formulas (just ~ 100 variables) that are completely beyond reach of even the very best SAT solvers
- Best known SAT solving algorithms based on Davis-Putnam-Logemann-Loveland or DPLL method from early 1960s (although with many clever optimizations)

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems
- But... There are also small formulas (just ~ 100 variables) that are completely beyond reach of even the very best SAT solvers
- Best known SAT solving algorithms based on Davis-Putnam-Logemann-Loveland or DPLL method from early 1960s (although with many clever optimizations)
- Some natural questions:
 - How do these SAT solvers work?

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems
- But... There are also small formulas (just ~ 100 variables) that are completely beyond reach of even the very best SAT solvers
- Best known SAT solving algorithms based on Davis-Putnam-Logemann-Loveland or DPLL method from early 1960s (although with many clever optimizations)
- Some natural questions:
 - How do these SAT solvers work?
 - How can they be so good in practice?

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems
- But... There are also small formulas (just ~ 100 variables) that are completely beyond reach of even the very best SAT solvers
- Best known SAT solving algorithms based on Davis-Putnam-Logemann-Loveland or DPLL method from early 1960s (although with many clever optimizations)
- Some natural questions:
 - How do these SAT solvers work?
 - How can they be so good in practice?
 - When they fail to be efficient, can we understand why?

Solving Logic Formulas in Practice

- Enormous progress since the turn of the millennium on so-called SAT solvers
Today routinely used to solve large-scale real-world problems
- But... There are also small formulas (just ~ 100 variables) that are completely beyond reach of even the very best SAT solvers
- Best known SAT solving algorithms based on Davis-Putnam-Logemann-Loveland or DPLL method from early 1960s (although with many clever optimizations)
- Some natural questions:
 - How do these SAT solvers work?
 - How can they be so good in practice?
 - When they fail to be efficient, can we understand why?
 - It's 2024 now — can we go beyond techniques from 1960s?

Plan for Today

What we will cover today:

Plan for Today

What we will cover today:

- Define more precisely the computational problem

Plan for Today

What we will cover today:

- Define more precisely the computational problem
- Give (slightly simplified) description of how modern SAT solvers work

Plan for Today

What we will cover today:

- Define more precisely the computational problem
- Give (slightly simplified) description of how modern SAT solvers work
- Present tools to analyze SAT solver performance

Plan for Today

What we will cover today:

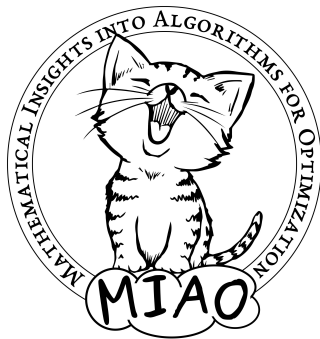
- Define more precisely the computational problem
- Give (slightly simplified) description of how modern SAT solvers work
- Present tools to analyze SAT solver performance
- Discuss how to extend SAT techniques to 0–1 integer linear programs and beyond
[in the pseudo-Boolean tutorials]

Plan for Today

What we will cover today:

- Define more precisely the computational problem
- Give (slightly simplified) description of how modern SAT solvers work
- Present tools to analyze SAT solver performance
- Discuss how to extend SAT techniques to 0–1 integer linear programs and beyond
[in the pseudo-Boolean tutorials]

... And in the process also touch on some of the research conducted in the *Mathematical Insights into Algorithms for Optimization (MIAO)* group in Copenhagen and Lund



Outline of Tutorial on Boolean Satisfiability (SAT) Solving

1 Boolean Satisfiability (SAT) Solving

- The SATISFIABILITY Problem
- Davis-Putnam-Logemann-Loveland (DPLL) Method
- Conflict-Driven Clause Learning (CDCL)

2 Proof Complexity

- Resolution Proof System
- Resolution and SAT Solving
- Lower Bounds for Resolution

3 Future Research Directions

- Understanding and Improving on the State of the Art
- Pseudo-Boolean Solving and the Cutting Planes Method
- Some Research Questions

Formal Description of SAT Problem

- **Variable** x : takes value 1 (**true**) or 0 (**false**)
- **Literal** ℓ : variable x or its negation $\bar{x}$ (write $\bar{x}$ instead of $\neg x$)
- **Clause** $C = \ell_1 \vee \dots \vee \ell_k$: disjunction of literals
(Consider as sets, so no repetitions and order irrelevant)
- **Conjunctive normal form (CNF) formula** $F = C_1 \wedge \dots \wedge C_m$: conjunction of clauses

Formal Description of SAT Problem

- **Variable** x : takes value 1 (**true**) or 0 (**false**)
- **Literal** ℓ : variable x or its negation $\bar{x}$ (write $\bar{x}$ instead of $\neg x$)
- **Clause** $C = \ell_1 \vee \dots \vee \ell_k$: disjunction of literals
(Consider as sets, so no repetitions and order irrelevant)
- **Conjunctive normal form (CNF) formula** $F = C_1 \wedge \dots \wedge C_m$: conjunction of clauses

The SATISFIABILITY (or just SAT) Problem

Given a CNF formula F , is it satisfiable?

Formal Description of SAT Problem

- **Variable** x : takes value 1 (**true**) or 0 (**false**)
- **Literal** ℓ : variable x or its negation $\bar{x}$ (write $\bar{x}$ instead of $\neg x$)
- **Clause** $C = \ell_1 \vee \dots \vee \ell_k$: disjunction of literals
 (Consider as sets, so no repetitions and order irrelevant)
- **Conjunctive normal form (CNF) formula** $F = C_1 \wedge \dots \wedge C_m$: conjunction of clauses

The SATISFIABILITY (or just SAT) Problem

Given a CNF formula F , is it satisfiable?

For instance, what about our example formula?

$$\begin{aligned} & (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ & \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w}) \end{aligned}$$

How to Solve the SAT Problem?

- Let computer check all possible assignments! Isn't this exactly the kind of monotone routine work at which computers excel?
- But how many cases to check?
- Suppose formula has n variables
- Each variable can be either true or false, so all in all get 2^n different cases
- If formula contains, say, one million variables, we get $2^{1,000,000}$ cases (a number with more than 300,000 digits)

How to Solve the SAT Problem?

- Let computer check all possible assignments! Isn't this exactly the kind of monotone routine work at which computers excel?
- But how many cases to check?
- Suppose formula has n variables
- Each variable can be either true or false, so all in all get 2^n different cases
- If formula contains, say, one million variables, we get $2^{1,000,000}$ cases (a number with more than 300,000 digits)

To understand how large this number is, consider that even if every atom in the known universe was a modern supercomputer that had been running at full speed ever since the beginning of time some 13.7 billion years ago, all of them together would only have covered a completely negligible fraction of these cases by now. So we really would not have time to wait for them to finish. . .

An Interesting Feature of the SAT Problem

- Deciding whether a satisfying assignment exists may take a long time
- But if you happen to know a satisfying assignment, easy to convince someone else that formula is satisfiable
- How? Just give assignment — can be verified in linear time
- So SAT problem **might seem hard to solve**, but **verifying a solution is easy** (not all problems have this property — how do you verify a winning position in chess?)
- The family of problems for which solutions are easy to check have a name: **NP**

How to Solve the SAT Problem, Take 2

- SAT problem can be used to describe any problem in NP — it is **NP-complete** [Coo71, Lev73]
- If you can solve SAT efficiently, then you can solve any problem in NP efficiently (this is why SAT is so useful)
- So **how hard is it to solve SAT?** (Brute force didn't work, but it usually doesn't — maybe can do something smarter?)

How to Solve the SAT Problem, Take 2

- SAT problem can be used to describe any problem in NP — it is **NP-complete** [Coo71, Lev73]
- If you can solve SAT efficiently, then you can solve any problem in NP efficiently (this is why SAT is so useful)
- So **how hard is it to solve SAT?** (Brute force didn't work, but it usually doesn't — maybe can do something smarter?)
- We don't know
- This one of the million-dollar “**Millennium Prize Problems**” [Mil00] posed as key challenges for mathematics in the new millennium
- Widely believe to be impossible to solve efficiently on computer in the worst case, but we really don't know

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the [DPLL method](#) developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the **DPLL method** developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

DPLL (somewhat simplified description)

- If F contains empty clause (without literals), report “**unsatisfiable**” and return — refer to as **conflict**

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the **DPLL method** developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

DPLL (somewhat simplified description)

- If F contains empty clause (without literals), report “**unsatisfiable**” and return — refer to as **conflict**
- If F contains no clauses, report “**satisfiable**” and terminate

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the **DPLL method** developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

DPLL (somewhat simplified description)

- If F contains empty clause (without literals), report “**unsatisfiable**” and return — refer to as **conflict**
- If F contains no clauses, report “**satisfiable**” and terminate
- Otherwise pick some variable x in F

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the **DPLL method** developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

DPLL (somewhat simplified description)

- If F contains empty clause (without literals), report “**unsatisfiable**” and return — refer to as **conflict**
- If F contains no clauses, report “**satisfiable**” and terminate
- Otherwise pick some variable x in F
- Set $x = 0$, simplify F and **make recursive call**

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the **DPLL method** developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

DPLL (somewhat simplified description)

- If F contains empty clause (without literals), report “**unsatisfiable**” and return — refer to as **conflict**
- If F contains no clauses, report “**satisfiable**” and terminate
- Otherwise pick some variable x in F
- Set $x = 0$, simplify F and **make recursive call**
- Set $x = 1$, simplify F and **make recursive call**

An Attempt at a Smarter Case Analysis: DPLL

Ok, but suppose you're out there in reality and actually have to solve the problem — then what do you do?

Chances are you'll use some variant of the **DPLL method** developed by Davis, Putnam, Logemann & Loveland [DP60, DLL62]

DPLL (somewhat simplified description)

- If F contains empty clause (without literals), report “**unsatisfiable**” and return — refer to as **conflict**
- If F contains no clauses, report “**satisfiable**” and terminate
- Otherwise pick some variable x in F
- Set $x = 0$, simplify F and **make recursive call**
- Set $x = 1$, simplify F and **make recursive call**
- If result in both cases “**unsatisfiable**”, then report “**unsatisfiable**” and return

A DPLL Toy Example

$$\begin{aligned} F = & (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ & \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w}) \end{aligned}$$

A DPLL Toy Example

$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals

A DPLL Toy Example

$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

$$F = (z) \wedge (y \vee \bar{z}) \wedge (\bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

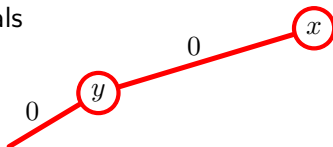
$$F = (z) \wedge (\bar{z}) \wedge (\bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

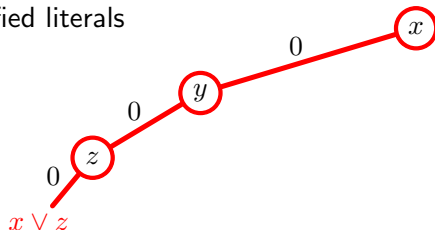
$$F = (x \vee z) \wedge (\bar{z}) \wedge (\bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

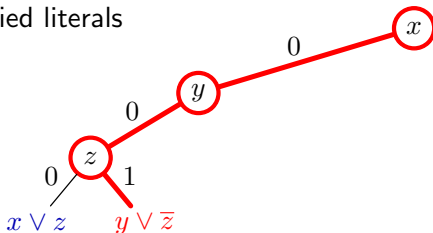
$$F = (z \wedge (y \vee \bar{z}) \wedge (\bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

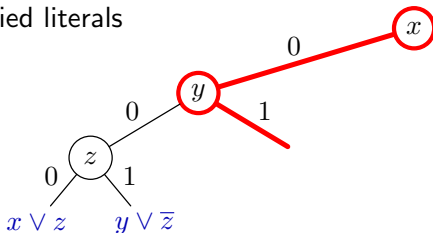
$$F = (z) \wedge (y \vee \bar{z}) \wedge (u) \wedge (\bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

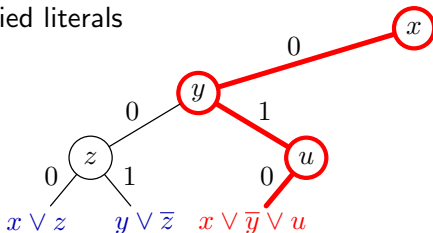
$$F = (z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{u}) \\ \wedge (v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

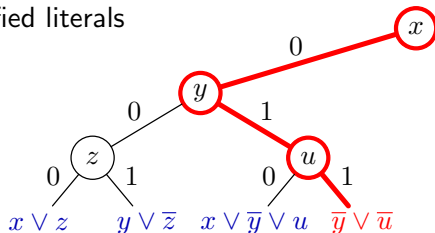
$$F = (z) \wedge (y \vee \bar{z}) \wedge (u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (w) \wedge (\bar{x} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

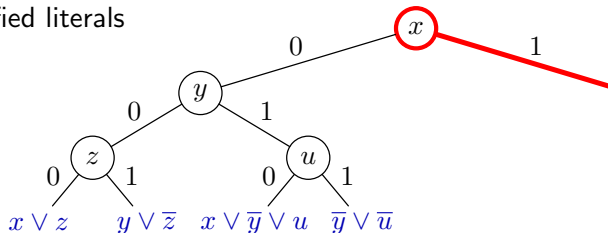
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

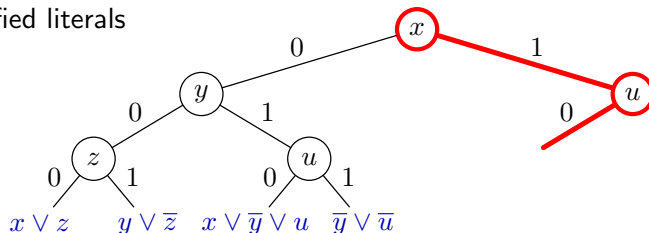
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y}) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (v) \wedge (\bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

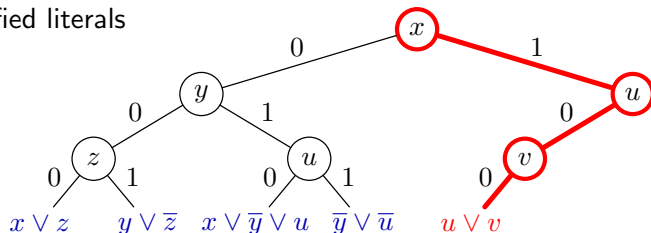
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y}) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

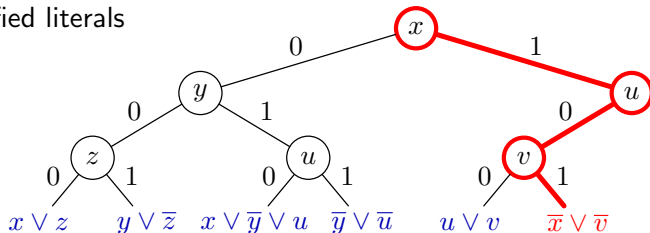
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y}) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

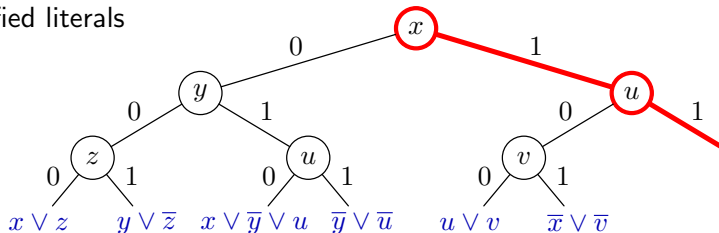
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{v}) \wedge (w) \wedge (\bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

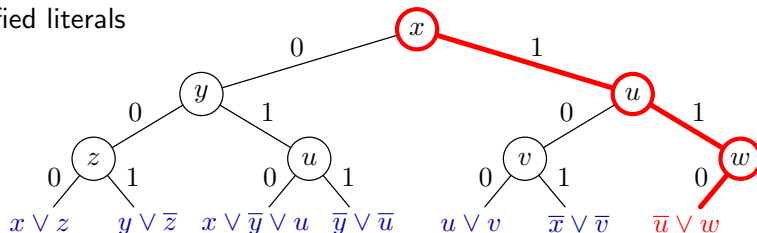
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

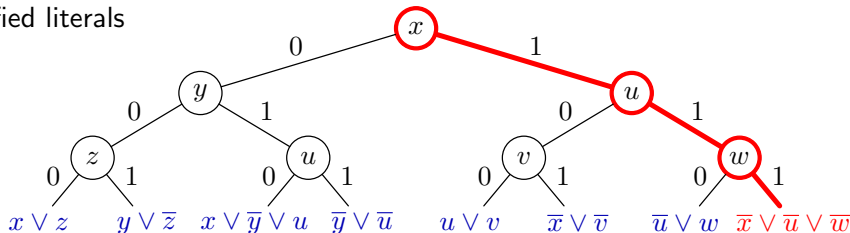
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{v}) \wedge (w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



A DPLL Toy Example

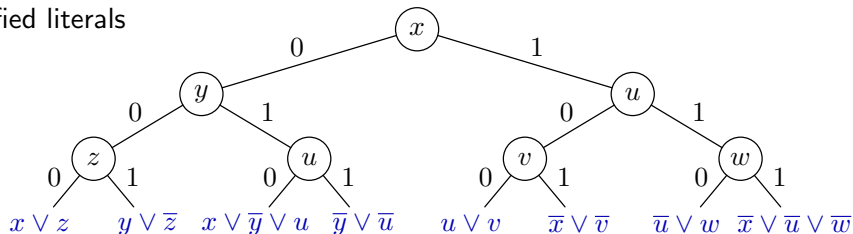
$$F = (x \vee z) \wedge (y \vee \bar{z}) \wedge (x \vee \bar{y} \vee u) \wedge (\bar{y} \vee \bar{u}) \\ \wedge (u \vee v) \wedge (\bar{x} \vee \bar{v}) \wedge (\bar{u} \vee w) \wedge (\bar{x} \vee \bar{u} \vee \bar{w})$$

Visualize execution of DPLL algorithm as search tree

Pick variables in internal nodes; terminate in leaves when conflict reached

“Simplify formula” by (mentally) removing

- satisfied clauses
- falsified literals



State-of-the-art SAT Solvers: Ingredients

Many more ingredients in modern **conflict-driven clause learning (CDCL)** SAT solvers (as pioneered in [BS97, MS99, MMZ⁺01]), e.g.:

- **Branching** or **decision heuristic** (choice of pivot variables crucial)
- When reaching leaf, **compute explanation for conflict** and **add to formula** as new clause (**clause learning**)
- Every once in a while, **restart** from beginning (but save computed info)
- **Preprocessing** the formula before the search even starts

Let us discuss some of these ingredients

Variable Assignment Heuristics

Unit propagation

- Suppose current assignment ρ falsifies all literals in $C = \ell_1 \vee \ell_2 \vee \dots \vee \ell_k$ except one (say ℓ_k) — C is **unit under ρ**
- Then ℓ_k has to be true, so set it to true
- Known as **unit propagation** or **Boolean constraint propagation**
- Always propagate if possible — in modern solvers aim for $\approx 99\%$ of assignments being unit propagations

Variable Assignment Heuristics

Unit propagation

- Suppose current assignment ρ falsifies all literals in $C = \ell_1 \vee \ell_2 \vee \dots \vee \ell_k$ except one (say ℓ_k) — C is **unit under ρ**
- Then ℓ_k has to be true, so set it to true
- Known as **unit propagation** or **Boolean constraint propagation**
- Always propagate if possible — in modern solvers aim for $\approx 99\%$ of assignments being unit propagations

VSIDS (Variable state independent decaying sum)

- When backtracking, score $+1$ for variables “causing conflict”
- Also multiply all scores with factor $\kappa < 1$ — exponential filter rewarding variables involved in recent conflicts
- When no propagations, **decide** on variable with highest score

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again
- Suppose we can compute that **decisions** $x = 1, y = 0, z = 1$ responsible for conflict

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again
- Suppose we can compute that **decisions** $x = 1, y = 0, z = 1$ responsible for conflict
- Then can add $\bar{x} \vee y \vee \bar{z}$ to avoid these decisions being made again — **decision learning scheme** (this idea goes all the way back to [SS77])

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again
- Suppose we can compute that **decisions** $x = 1, y = 0, z = 1$ responsible for conflict
- Then can add $\bar{x} \vee y \vee \bar{z}$ to avoid these decisions being made again — **decision learning scheme** (this idea goes all the way back to [SS77])
- Nowadays, more sophisticated learning schemes starting with [MS99, MMZ⁺01]

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again
- Suppose we can compute that **decisions** $x = 1, y = 0, z = 1$ responsible for conflict
- Then can add $\bar{x} \vee y \vee \bar{z}$ to avoid these decisions being made again — **decision learning scheme** (this idea goes all the way back to [SS77])
- Nowadays, more sophisticated learning schemes starting with [MS99, MMZ⁺01]
- Often described in terms of cuts in **conflict graph**

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again
- Suppose we can compute that **decisions** $x = 1, y = 0, z = 1$ responsible for conflict
- Then can add $\bar{x} \vee y \vee \bar{z}$ to avoid these decisions being made again — **decision learning scheme** (this idea goes all the way back to [SS77])
- Nowadays, more sophisticated learning schemes starting with [MS99, MMZ⁺01]
- Often described in terms of cuts in **conflict graph**
- More helpful to view conflict analysis as **syntactic derivation** applied on clauses unit propagating to conflict

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$\boxed{p \stackrel{d}{=} 0}$$

$$\boxed{u \stackrel{p \vee \bar{u}}{=} 0}$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{\perp}{=}$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{\perp}{=}$$

decision
level 1

decision
level 2

decision
level 3

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

Always propagate if possible, otherwise decide

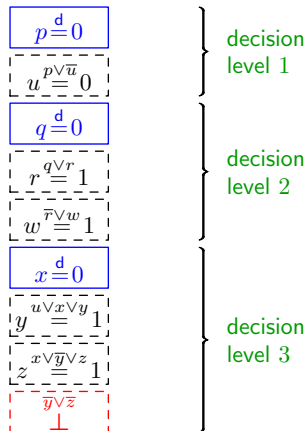
Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{\perp}{=}$$

decision
level 1

decision
level 2

decision
level 3

Could backtrack by erasing **conflict level** & flipping last decision

Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{\perp}{=}$$

decision
level 1

decision
level 2

decision
level 3

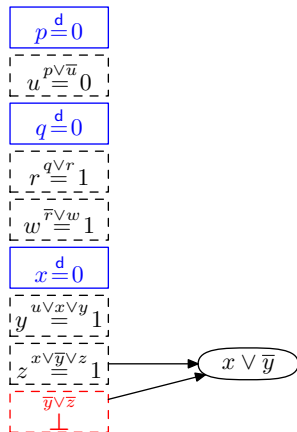
Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

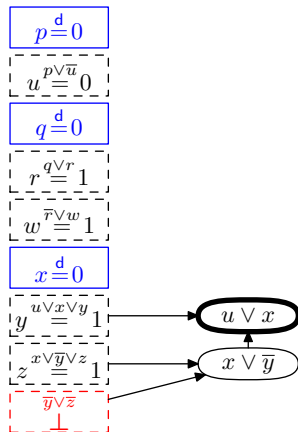
Case analysis for last two clauses over propagated variable:

- $x \vee \bar{y} \vee z$ wants $z = 1$
- $\bar{y} \vee \bar{z}$ wants $z = 0$
- Merge clauses & remove z — must satisfy $x \vee \bar{y}$

Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

Case analysis for last two clauses over propagated variable:

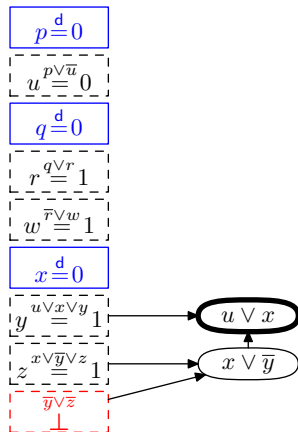
- $x \vee \bar{y} \vee z$ wants $z = 1$
- $\bar{y} \vee \bar{z}$ wants $z = 0$
- Merge clauses & remove z — must satisfy $x \vee \bar{y}$

Repeat until **UIP clause** with only 1 variable at conflict level after last decision — **learn** and **backjump**

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

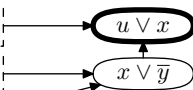
$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \perp$$



$$p \stackrel{d}{=} 0$$

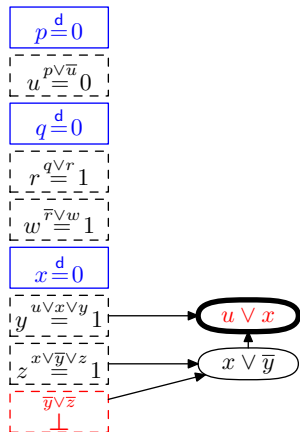
$$u \stackrel{p \vee \bar{u}}{=} 0$$

Assertion level 1 (2nd largest level in learned clause) —
trim trail to that level

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



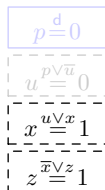
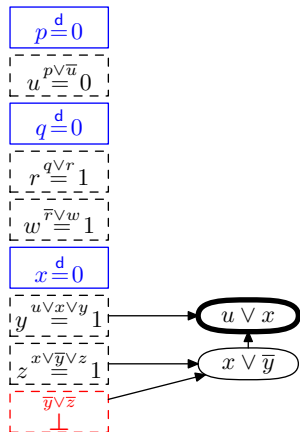
Assertion level 1 (2nd largest level in learned clause) — trim trail to that level

Now UIP literal guaranteed to flip (**assert**) — but this is a **propagation**, not a decision

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Assertion level 1 (2nd largest level in learned clause) —
trim trail to that level

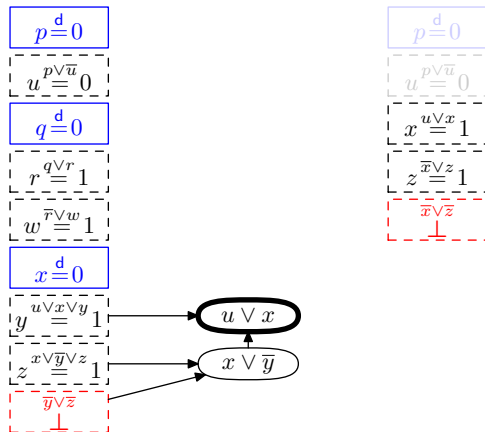
Now UIP literal guaranteed to flip (**assert**) — but this is a **propagation**, not a decision

Then continue as before. . .

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

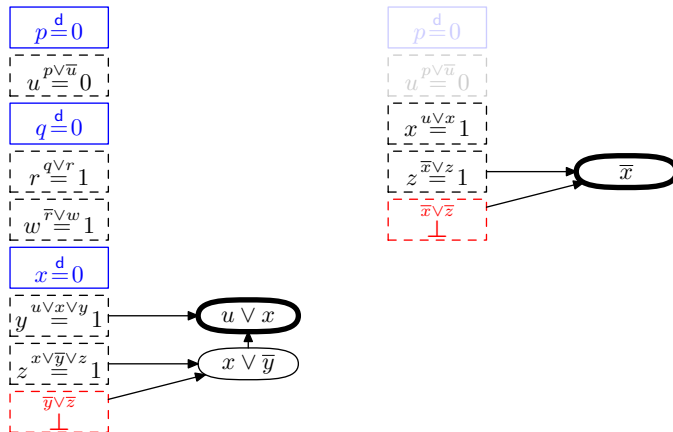
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

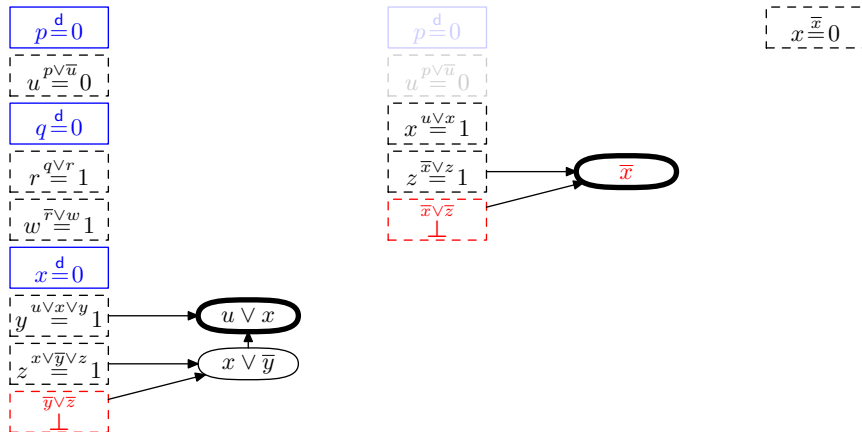
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

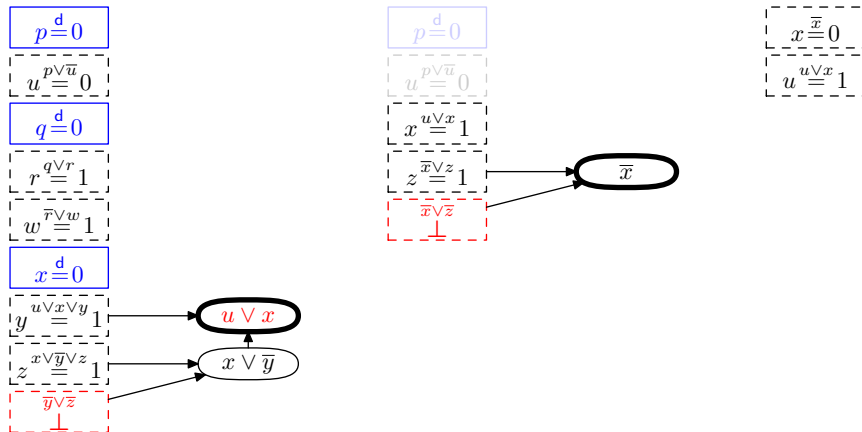
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

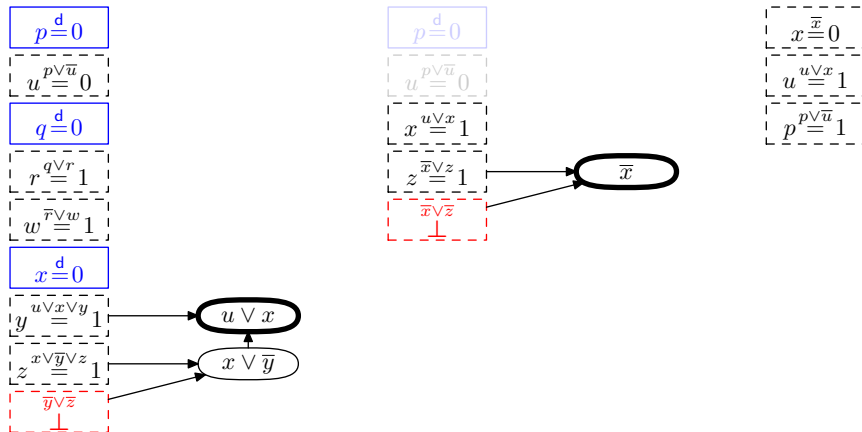
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

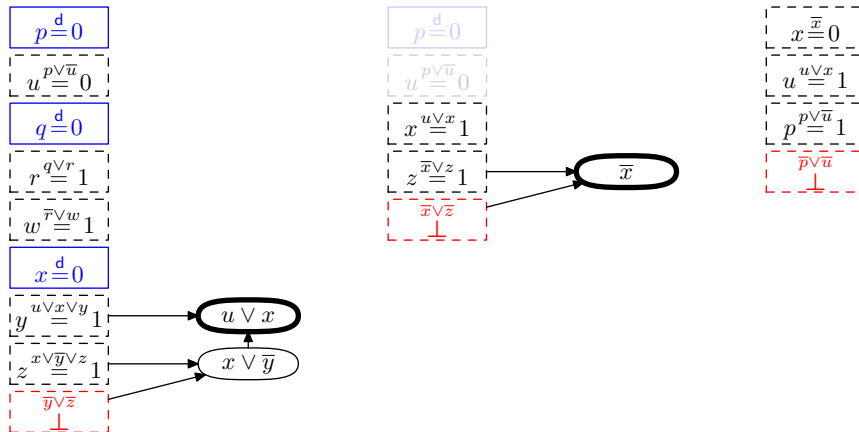
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

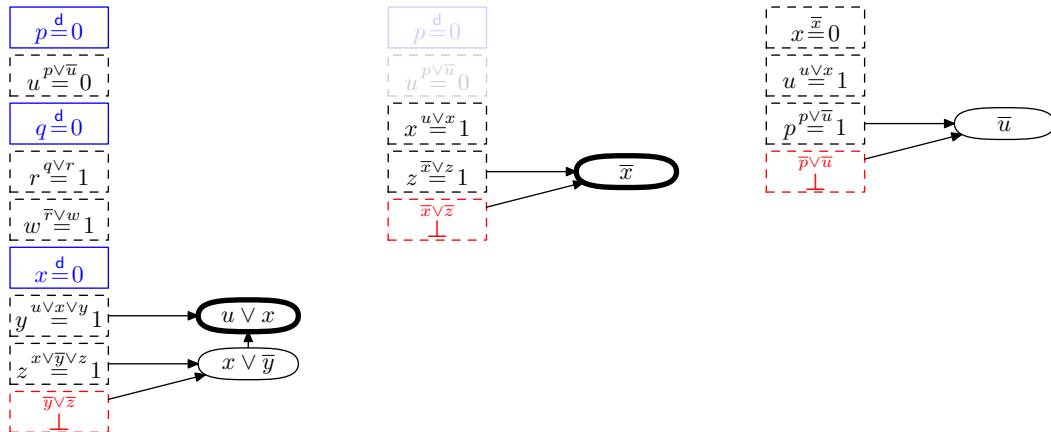
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

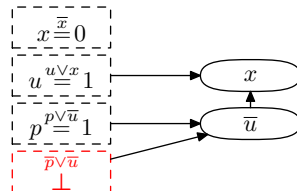
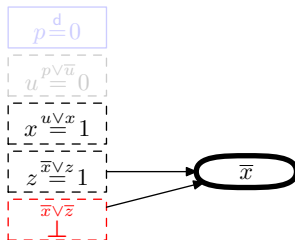
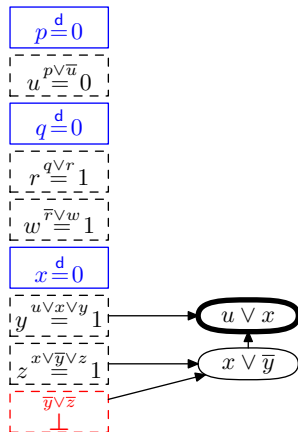
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

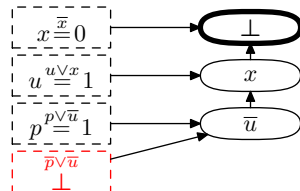
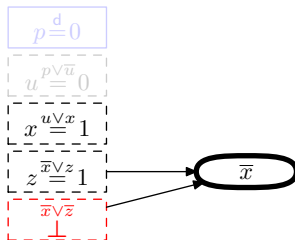
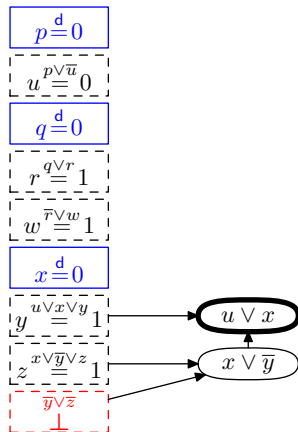
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Clause Database Reduction

- In addition to learning clauses, also **erase learned clauses that don't seem useful**
- Modern solvers do this **very aggressively**
- Speeds up CDCL search (in particular, unit propagation, which dominates running time)
- But erasing too aggressively can throw away clauses that would have made solver terminate faster [EGG⁺18]
- So potential **trade-off** between **search speed** and **search quality**
- Except sometimes getting rid of clauses improves search quality too! [KN20]

Restarts

- Fairly frequently, **start search all over** (but keep learned clauses)
- Original intuition: stuck in bad part of search tree — go somewhere else
- Not the reason this is done now
- Popular variables with high VSIDS scores get set again [MMZ⁺01]
- Are even set to same values (**phase saving**) [PD07]
- Current intuition: improves the search by focusing on important variables
- Restart at fixed intervals or (better) make **adaptive restarts** depending on “quality” of learned clauses [AS09, AS12]

Conflict-Driven Clause Learning in Pseudocode (Slightly Simplified)

CDCL(F)

```

1  $\mathcal{D} \leftarrow F$  ; // initialize clause database to contain formula
2  $\rho \leftarrow \emptyset$  ; // initialize assignment trail to empty
3 forever do
4   if  $\rho$  falsifies some clause  $C \in \mathcal{D}$  then
5      $A \leftarrow \text{analyzeConflict}(\mathcal{D}, \rho, C)$  ;
6     if  $A = \perp$  then output UNSATISFIABLE and exit ;
7     else add learned clause  $A$  to  $\mathcal{D}$  and backjump by shrinking  $\rho$  ;
8   else if exists clause  $C \in \mathcal{D}$  unit propagating  $x$  to  $b \in \{0, 1\}$  under  $\rho$  then
9     add propagated assignment  $x \stackrel{C}{=} b$  to  $\rho$  ;
10  else if time to restart then  $\rho \leftarrow \emptyset$  ;
11  else if time for clause database reduction then
12    erase (roughly) half of learned clauses in  $\mathcal{D} \setminus F$  from  $\mathcal{D}$ 
13  else if all variables assigned then output SATISFIABLE and exit ;
14  else
15    use decision scheme to choose  $x$  and  $b$  and add assignment  $x \stackrel{d}{=} b$  to  $\rho$  ;
    
```

Conflict-Driven Clause Learning in Pseudocode (Slightly Simplified)

CDCL(F)

```

1  $\mathcal{D} \leftarrow F$  ; // initialize clause database to contain formula
2  $\rho \leftarrow \emptyset$  ; // initialize assignment trail to empty
3 forever do
4   if  $\rho$  falsifies some clause  $C \in \mathcal{D}$  then
5      $A \leftarrow \text{analyzeConflict}(\mathcal{D}, \rho, C)$  ;
6     if  $A = \perp$  then output UNSATISFIABLE and exit ;
7     else add learned clause  $A$  to  $\mathcal{D}$  and backjump by shrinking  $\rho$  ;
8   else if exists clause  $C \in \mathcal{D}$  unit propagating  $x$  to  $b \in \{0, 1\}$  under  $\rho$  then
9     add propagated assignment  $x \stackrel{C}{=} b$  to  $\rho$  ;
10  else if time to restart then  $\rho \leftarrow \emptyset$  ;
11  else if time for clause database reduction then
12    | erase (roughly) half of learned clauses in  $\mathcal{D} \setminus F$  from  $\mathcal{D}$ 
13  else if all variables assigned then output SATISFIABLE and exit ;
14  else
15    | use decision scheme to choose  $x$  and  $b$  and add assignment  $x \stackrel{d}{=} b$  to  $\rho$  ;

```

Conflict Analysis Pseudocode

$\text{analyzeConflict}(\mathcal{D}, \rho, C_{\text{confl}})$

```

1  $C_{\text{learn}} \leftarrow C_{\text{confl}} ;$ 
2 while  $C_{\text{learn}}$  not UIP clause and  $C_{\text{learn}} \neq \perp$  do
3    $\ell \leftarrow$  literal assigned last on trail  $\rho ;$ 
4   if  $\ell$  propagated and  $\bar{\ell}$  occurs in  $C_{\text{learn}}$  then
5      $C_{\text{reason}} \leftarrow \text{reason}(\ell, \rho, \mathcal{D}) ;$ 
6      $C_{\text{learn}} \leftarrow \text{resolve}(C_{\text{learn}}, C_{\text{reason}}) ;$ 
7    $\rho \leftarrow \rho \setminus \{\ell\} ;$ 
8 return  $C_{\text{learn}} ;$ 
    
```

State-of-the-art SAT solvers: What About the Recipe?

List of ingredients again (not exhaustive):

- Variable decisions & propagations
- Conflict analysis
- Restarts
- Clause database reduction
- Preprocessing

State-of-the-art SAT solvers: What About the Recipe?

List of ingredients again (not exhaustive):

- Variable decisions & propagations
- Conflict analysis
- Restarts
- Clause database reduction
- Preprocessing

Some natural questions:

- How best to combine these ingredients into a recipe?
- When and why does this recipe work?

State-of-the-art SAT solvers: What About the Recipe?

List of ingredients again (not exhaustive):

- Variable decisions & propagations
- Conflict analysis
- Restarts
- Clause database reduction
- Preprocessing

Some natural questions:

- How best to combine these ingredients into a recipe?
- When and why does this recipe work?

Why SAT solvers actually work so well is poorly understood

Plenty of research has been done to comprehend this better
(*Among other places in the MIAO group*)



SAT Solver Analysis and the Resolution Proof System

How to make **rigorous** analysis of CDCL SAT solver performance?

Many intricate, hard-to-understand heuristics

So focus instead on **underlying method of reasoning**

SAT Solver Analysis and the Resolution Proof System

How to make **rigorous** analysis of CDCL SAT solver performance?

Many intricate, hard-to-understand heuristics

So focus instead on **underlying method of reasoning**

Resolution proof system [Bla37, Rob65]

- Start with clauses of CNF formula (**axioms**)
- Derive new clauses by **resolution rule**

$$\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$$

Resolution Proofs by Contradiction

Resolution rule:

$$\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$$

Observation

If F is a satisfiable CNF formula and D is derived from clauses $D_1, D_2 \in F$ by the resolution rule, then $F \wedge D$ is satisfiable.

Resolution Proofs by Contradiction

Resolution rule:

$$\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$$

Observation

If F is a satisfiable CNF formula and D is derived from clauses $D_1, D_2 \in F$ by the resolution rule, then $F \wedge D$ is satisfiable.

So can prove F **unsatisfiable** by deriving the unsatisfiable empty clause (denoted $\perp$) from F by resolution

Resolution Proofs by Contradiction

Resolution rule:

$$\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$$

Observation

If F is a satisfiable CNF formula and D is derived from clauses $D_1, D_2 \in F$ by the resolution rule, then $F \wedge D$ is satisfiable.

So can prove F **unsatisfiable** by deriving the unsatisfiable empty clause (denoted $\perp$) from F by resolution

Such proof by contradiction also called **resolution refutation**

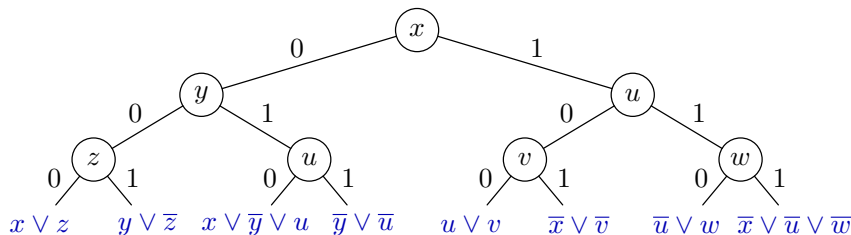
DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

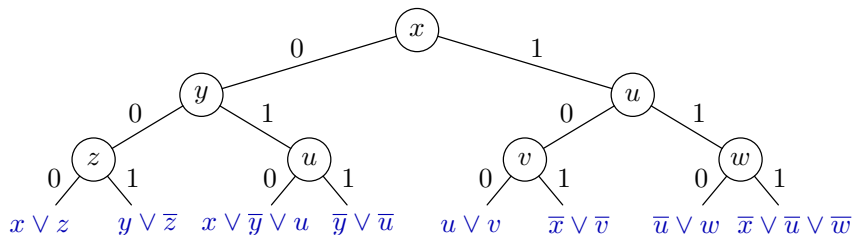
Look at our example again



DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

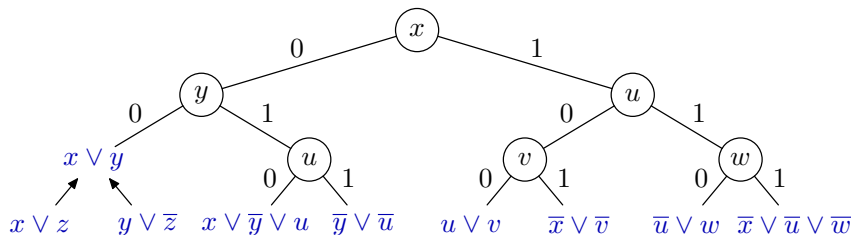


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

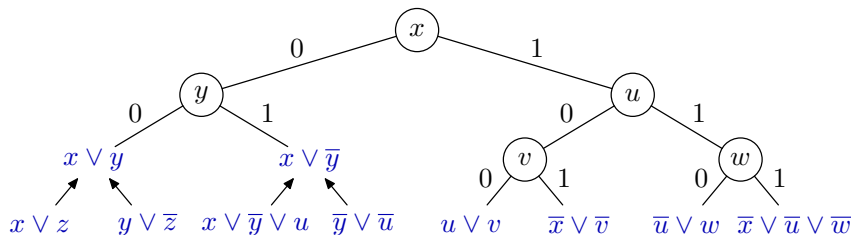


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

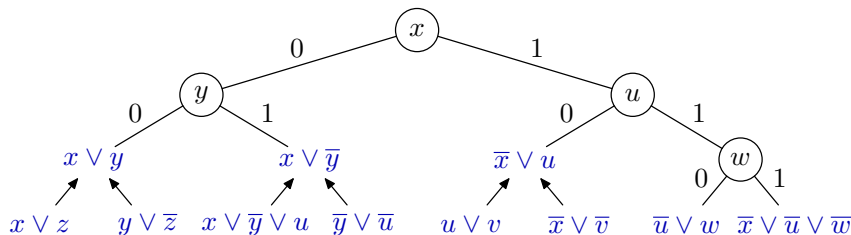


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

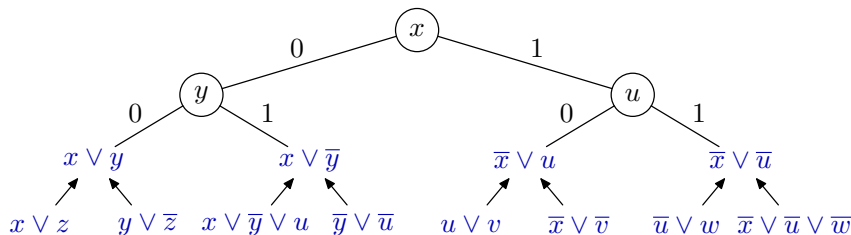


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

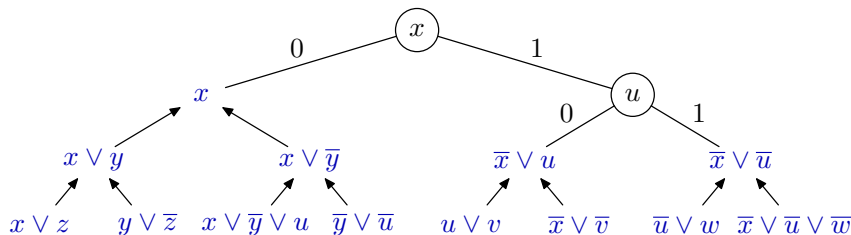


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

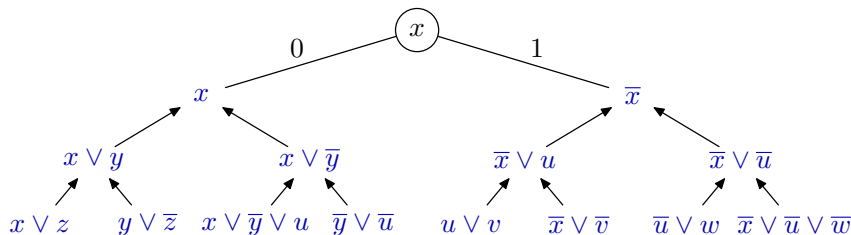


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again

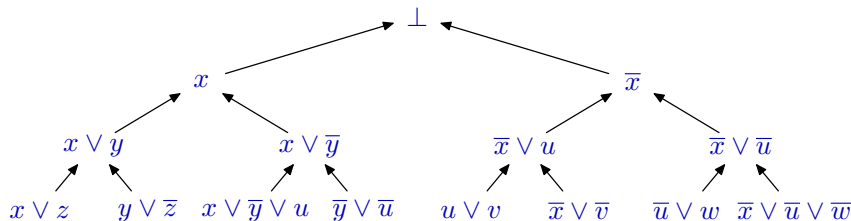


and **apply resolution rule** $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ **bottom-up**

DPLL and Resolution Proofs

A DPLL execution is essentially a resolution proof

Look at our example again



and apply resolution rule $\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$ bottom-up

DPLL Running Time and Tree-Like Resolution Proof Size

- Can extract resolution proof from any DPLL execution

DPLL Running Time and Tree-Like Resolution Proof Size

- Can extract resolution proof from any DPLL execution
- Requires an argument, of course, but not too hard to show

DPLL Running Time and Tree-Like Resolution Proof Size

- Can extract resolution proof from any DPLL execution
- Requires an argument, of course, but not too hard to show
- Such proof is **tree-like** — every derived clause **used only once**
(to use a clause twice, we have to derive it twice from scratch)

DPLL Running Time and Tree-Like Resolution Proof Size

- Can extract resolution proof from any DPLL execution
- Requires an argument, of course, but not too hard to show
- Such proof is **tree-like** — every derived clause **used only once** (to use a clause twice, we have to derive it twice from scratch)
- Hence, **lower bounds** on **tree-like proof size** in resolution $\Rightarrow$ lower bounds on **DPLL running time**

DPLL Running Time and Tree-Like Resolution Proof Size

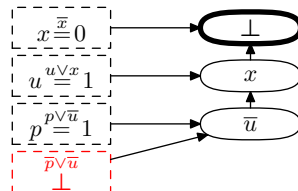
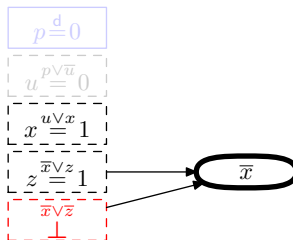
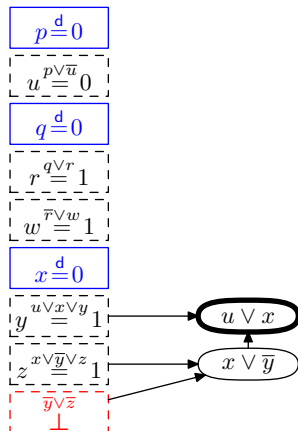
- Can extract resolution proof from any DPLL execution
- Requires an argument, of course, but not too hard to show
- Such proof is **tree-like** — every derived clause **used only once** (to use a clause twice, we have to derive it twice from scratch)
- Hence, **lower bounds** on **tree-like proof size** in resolution $\Rightarrow$ lower bounds on **DPLL running time**
- Conflict-driven clause learning adds “shortcut edges” in tree, but still yields resolution proof

CDCL and Resolution Proofs

Obtain resolution proof. . .

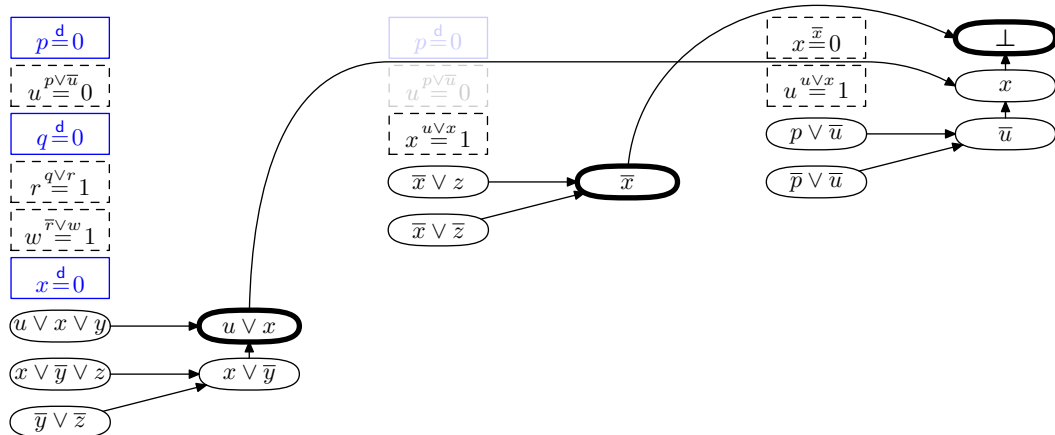
CDCL and Resolution Proofs

Obtain resolution proof from our example CDCL execution...



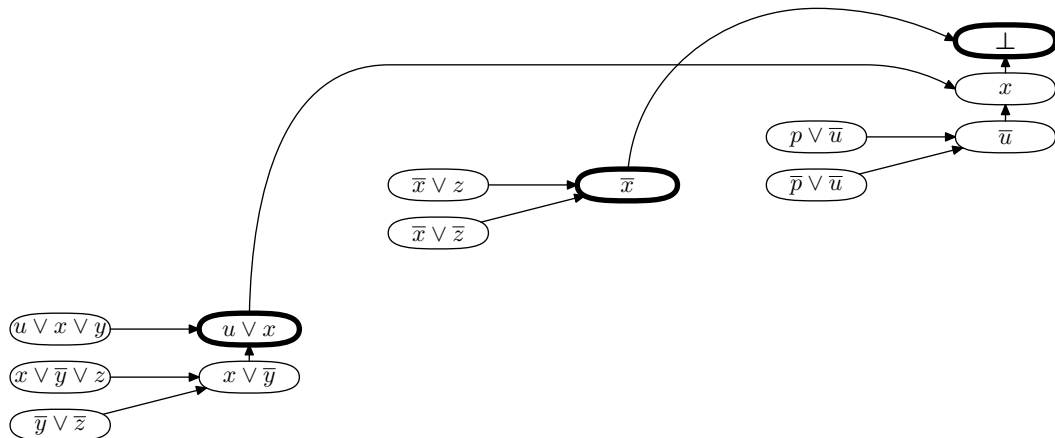
CDCL and Resolution Proofs

Obtain resolution proof from our example CDCL execution by stringing together conflict analyses:



CDCL and Resolution Proofs

Obtain resolution proof from our example CDCL execution by stringing together conflict analyses:



CDCL Running Time and General Resolution Proof Size

- Can extract general resolution proof (DAG-like, not tree-like) from CDCL execution

CDCL Running Time and General Resolution Proof Size

- Can extract general resolution proof (DAG-like, not tree-like) from CDCL execution
- Again requires an argument, but you have seen enough in this lecture to be able to fill in the required details...

CDCL Running Time and General Resolution Proof Size

- Can extract general resolution proof (DAG-like, not tree-like) from CDCL execution
- Again requires an argument, but you have seen enough in this lecture to be able to fill in the required details...
- This holds even for CDCL solvers with sophisticated heuristics and optimizations that we have not discussed*

CDCL Running Time and General Resolution Proof Size

- Can extract general resolution proof (DAG-like, not tree-like) from CDCL execution
- Again requires an argument, but you have seen enough in this lecture to be able to fill in the required details...
- This holds even for CDCL solvers with sophisticated heuristics and optimizations that we have not discussed*
- Hence, **lower bounds on resolution proof size** $\Rightarrow$
lower bounds on **CDCL running time**

CDCL Running Time and General Resolution Proof Size

- Can extract general resolution proof (DAG-like, not tree-like) from CDCL execution
- Again requires an argument, but you have seen enough in this lecture to be able to fill in the required details...
- This holds even for CDCL solvers with sophisticated heuristics and optimizations that we have not discussed*
- Hence, **lower bounds** on **resolution proof size** $\Rightarrow$
lower bounds on **CDCL running time**
- Lower (and upper) bounds for different methods of reasoning about propositional logic formulas studied in **proof complexity**

CDCL Running Time and General Resolution Proof Size

- Can extract general resolution proof (DAG-like, not tree-like) from CDCL execution
- Again requires an argument, but you have seen enough in this lecture to be able to fill in the required details. . .
- This holds even for CDCL solvers with sophisticated heuristics and optimizations that we have not discussed*
- Hence, **lower bounds** on **resolution proof size** $\Rightarrow$ lower bounds on **CDCL running time**
- Lower (and upper) bounds for different methods of reasoning about propositional logic formulas studied in **proof complexity**

(*) Except for some **preprocessing techniques**, which is an important omission, but this gets complicated and we don't have time to go into details. . .

Current State of Affairs in SAT Solving

- State-of-the-art CDCL solvers often perform amazingly well
(“SAT is easy in practice”)

Current State of Affairs in SAT Solving

- State-of-the-art CDCL solvers often perform amazingly well
(“SAT is easy in practice”)
- Very poor theoretical understanding:
 - Why do heuristics work?
 - Why are applied instances easy?

Current State of Affairs in SAT Solving

- State-of-the-art CDCL solvers often perform amazingly well
(“SAT is easy in practice”)
- Very poor theoretical understanding:
 - Why do heuristics work?
 - Why are applied instances easy?
- Paradox: resolution quite weak proof system; many strong proof complexity lower bounds for (seemingly) “obvious” formulas

Examples of Hard Formulas for Resolution (1/3)

Pigeonhole principle (PHP) formulas [Hak85]

“ $n + 1$ pigeons don't fit into n holes”

Examples of Hard Formulas for Resolution (1/3)

Pigeonhole principle (PHP) formulas [Hak85]

“ $n + 1$ pigeons don't fit into n holes”

Variables $p_{i,j}$ = “pigeon $i \rightarrow$ hole j ”; $1 \leq i \leq n + 1$; $1 \leq j \leq n$

$$p_{i,1} \vee p_{i,2} \vee \cdots \vee p_{i,n}$$

every pigeon i gets a hole

$$\bar{p}_{i,j} \vee \bar{p}_{i',j}$$

no hole j gets two pigeons $i \neq i'$

Can also add “functionality” and “onto” axioms

$$\bar{p}_{i,j} \vee \bar{p}_{i,j'}$$

no pigeon i gets two holes $j \neq j'$

$$p_{1,j} \vee p_{2,j} \vee \cdots \vee p_{n+1,j}$$

every hole j gets a pigeon

Examples of Hard Formulas for Resolution (1/3)

Pigeonhole principle (PHP) formulas [Hak85]

“ $n + 1$ pigeons don't fit into n holes”

Variables $p_{i,j}$ = “pigeon $i \rightarrow$ hole j ”; $1 \leq i \leq n + 1$; $1 \leq j \leq n$

$$p_{i,1} \vee p_{i,2} \vee \cdots \vee p_{i,n}$$

every pigeon i gets a hole

$$\bar{p}_{i,j} \vee \bar{p}_{i',j}$$

no hole j gets two pigeons $i \neq i'$

Can also add “functionality” and “onto” axioms

$$\bar{p}_{i,j} \vee \bar{p}_{i,j'}$$

no pigeon i gets two holes $j \neq j'$

$$p_{1,j} \vee p_{2,j} \vee \cdots \vee p_{n+1,j}$$

every hole j gets a pigeon

Even onto functional PHP hard — **“resolution cannot count”**

Resolution proof requires $\exp(\Omega(n)) = \exp(\Omega(\sqrt[3]{N}))$ clauses
 (measured in terms of **formula size N**)

Examples of Hard Formulas for Resolution (2/3)

Tseitin formulas [Urq87]

“Sum of degrees of vertices in graph is even”

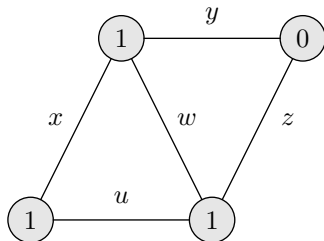
Examples of Hard Formulas for Resolution (2/3)

Tseitin formulas [Urq87]

“Sum of degrees of vertices in graph is even”

Variables = edges (in undirected graph of bounded degree)

- Label every vertex 0/1 so that sum of labels odd
- Write CNF requiring parity of $\#$ true incident edges = label



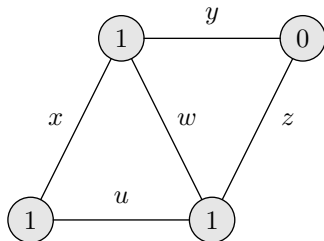
Examples of Hard Formulas for Resolution (2/3)

Tseitin formulas [Urq87]

“Sum of degrees of vertices in graph is even”

Variables = edges (in undirected graph of bounded degree)

- Label every vertex 0/1 so that sum of labels odd
- Write CNF requiring parity of $\#$ true incident edges = label



$$\begin{array}{ll}
 (u \vee x) & \wedge (y \vee \bar{z}) \\
 \wedge (\bar{u} \vee \bar{x}) & \wedge (\bar{y} \vee z) \\
 \wedge (w \vee x \vee y) & \wedge (u \vee w \vee z) \\
 \wedge (w \vee \bar{x} \vee \bar{y}) & \wedge (u \vee \bar{w} \vee \bar{z}) \\
 \wedge (\bar{w} \vee x \vee \bar{y}) & \wedge (\bar{u} \vee w \vee \bar{z}) \\
 \wedge (\bar{w} \vee \bar{x} \vee y) & \wedge (\bar{u} \vee \bar{w} \vee z)
 \end{array}$$

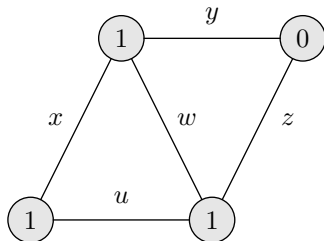
Examples of Hard Formulas for Resolution (2/3)

Tseitin formulas [Urq87]

“Sum of degrees of vertices in graph is even”

Variables = edges (in undirected graph of bounded degree)

- Label every vertex 0/1 so that sum of labels odd
- Write CNF requiring parity of $\#$ true incident edges = label



$$\begin{array}{ll}
 (u \vee x) & \wedge (y \vee \bar{z}) \\
 \wedge (\bar{u} \vee \bar{x}) & \wedge (\bar{y} \vee z) \\
 \wedge (w \vee x \vee y) & \wedge (u \vee w \vee z) \\
 \wedge (w \vee \bar{x} \vee \bar{y}) & \wedge (u \vee \bar{w} \vee \bar{z}) \\
 \wedge (\bar{w} \vee x \vee \bar{y}) & \wedge (\bar{u} \vee w \vee \bar{z}) \\
 \wedge (\bar{w} \vee \bar{x} \vee y) & \wedge (\bar{u} \vee \bar{w} \vee z)
 \end{array}$$

Requires **proof size** $\exp(\Omega(N))$ on well-connected so-called **expander graphs** —

“**resolution cannot count mod 2**”

Examples of Hard Formulas for Resolution (3/3)

Random k -CNF formulas [CS88]

Δn randomly sampled k -clauses over n variables

($\Delta \gtrsim 4.5$ sufficient to get unsatisfiable 3-CNF formula almost surely)

Again lower bound $\exp(\Omega(N))$

Examples of Hard Formulas for Resolution (3/3)

Random k -CNF formulas [CS88]

Δn randomly sampled k -clauses over n variables

($\Delta \gtrsim 4.5$ sufficient to get unsatisfiable 3-CNF formula almost surely)

Again lower bound $\exp(\Omega(N))$

And more...

- COLOURING [BCMM05]
- CLIQUE and VERTEXCOVER [BIS07] (though open questions remain [ABdR⁺21])
- Zero-one designs [Spe10, VS10, MN14]
- ...
- See Chapter 7 on *Proof Complexity and SAT Solving* in the *Handbook of Satisfiability* for more details [BN21]

Theoretical Lower Bounds and Practical Reality

- If resolution so weak, how can CDCL SAT solvers be so good?
- One answer: “tricky” formulas don’t show up too often in practice
- Another area of intense research: Try to describe **what properties of “real-life” formulas make them easy or hard**

Theoretical Lower Bounds and Practical Reality

- If resolution so weak, how can CDCL SAT solvers be so good?
- One answer: “tricky” formulas don’t show up too often in practice
- Another area of intense research: Try to describe **what properties of “real-life” formulas make them easy or hard**
- But sometimes we would like to be able to solve also “tricky” formulas
- Can we go beyond resolution?

Theoretical Lower Bounds and Practical Reality

- If resolution so weak, how can CDCL SAT solvers be so good?
- One answer: “tricky” formulas don’t show up too often in practice
- Another area of intense research: Try to describe **what properties of “real-life” formulas make them easy or hard**
- But sometimes we would like to be able to solve also “tricky” formulas
- Can we go beyond resolution?
- Explore **stronger methods of reasoning!**
- Algorithms based on such methods could potentially lead to **exponential speed-ups**
[stay tuned for following lectures...]

Cutting Planes Proof System

Introduced in [CCT87] to model integer linear programming algorithm in [Gom63, Chv73]

Clauses translated to **linear inequalities** over the reals with **integer coefficients**

Cutting Planes Proof System

Introduced in [CCT87] to model integer linear programming algorithm in [Gom63, Chv73]

Clauses translated to **linear inequalities** over the reals with **integer coefficients**

Example: $x \vee y \vee \bar{z}$ gets translated to $x + y + (1 - z) \geq 1$
 or equivalently $x + y - z \geq 0$

Derivation rules

Variable axioms
$$\frac{}{0 \leq x \leq 1}$$

Multiplication
$$\frac{\sum a_i x_i \geq A}{\sum c a_i x_i \geq c A} \quad [c \in \mathbb{N}^+]$$

Addition
$$\frac{\sum a_i x_i \geq A \quad \sum b_i x_i \geq B}{\sum (a_i + b_i) x_i \geq A + B}$$

Division
$$\frac{\sum c a_i x_i \geq A}{\sum a_i x_i \geq \lceil A/c \rceil} \quad [c \in \mathbb{N}^+]$$

Cutting Planes Refutations of CNF Formulas

- Translate CNF formula to set of 0–1 linear inequalities
- Apply derivation rules
- Derive $0 \geq 1 \Leftrightarrow$ formula unsatisfiable
- Also makes sense for more general 0–1 linear inequalities (not just translations of CNF formulas)
- Cutting planes can simulate resolution reasoning efficiently and is sometimes exponentially stronger (e.g., for PHP, just count to see $\#pigeons > \#holes$)

Building SAT Solvers Based on Cutting Planes Reasoning?

So-called **pseudo-Boolean solvers** using cutting planes developed in [CK05, LP10, EN18]
Counter-intuitively, **hard to make competitive with CDCL**

Building SAT Solvers Based on Cutting Planes Reasoning?

So-called **pseudo-Boolean solvers** using cutting planes developed in [CK05, LP10, EN18]
Counter-intuitively, **hard to make competitive with CDCL**

Challenge 1: Conjunctive normal form

- Pseudo-Boolean solvers terrible for CNF input
- Solvers can rewrite CNF to more helpful 0–1 linear inequalities [BLLM14, EN20], but hard to make this work well enough in practice
- Better to encode problem with 0–1 integer linear inequalities from the start

Building SAT Solvers Based on Cutting Planes Reasoning?

So-called **pseudo-Boolean solvers** using cutting planes developed in [CK05, LP10, EN18]
Counter-intuitively, **hard to make competitive with CDCL**

Challenge 1: Conjunctive normal form

- Pseudo-Boolean solvers terrible for CNF input
- Solvers can rewrite CNF to more helpful 0–1 linear inequalities [BLLM14, EN20], but hard to make this work well enough in practice
- Better to encode problem with 0–1 integer linear inequalities from the start

Challenge 2: Increased degrees of freedom(!?)

- Cutting planes much smarter method of reasoning
- But this makes it trickier to design smart search algorithms

Building SAT Solvers Based on Cutting Planes Reasoning?

So-called **pseudo-Boolean solvers** using cutting planes developed in [CK05, LP10, EN18]
Counter-intuitively, **hard to make competitive with CDCL**

Challenge 1: Conjunctive normal form

- Pseudo-Boolean solvers terrible for CNF input
- Solvers can rewrite CNF to more helpful 0–1 linear inequalities [BLLM14, EN20], but hard to make this work well enough in practice
- Better to encode problem with 0–1 integer linear inequalities from the start

Challenge 2: Increased degrees of freedom(!?)

- Cutting planes much smarter method of reasoning
- But this makes it trickier to design smart search algorithms

Is it truly harder to build good pseudo-Boolean solvers?

Or has just so much more work has been put into CDCL solvers?

So... Is There a Smarter Way Than Brute-Force to Solve SAT?

In theory, probably no...

- COLOURING, CLIQUE, SAT, and 1000s other problems are “all the same” — efficient algorithm for one can solve all (the problems are all **NP-complete**)
- Widely believed impossible to construct algorithms that are always (a) efficient and (b) correct (even in **worst case**)
- Settling this question is one of **Millennium Prize Problems**: Are there efficient algorithms for NP-complete problems?

So... Is There a Smarter Way Than Brute-Force to Solve SAT?

In theory, probably no...

- COLOURING, CLIQUE, SAT, and 1000s other problems are “all the same” — efficient algorithm for one can solve all (the problems are all **NP-complete**)
- Widely believed impossible to construct algorithms that are always (a) efficient and (b) correct (even in **worst case**)
- Settling this question is one of **Millennium Prize Problems**: Are there efficient algorithms for NP-complete problems?

In practice, definitely yes!

- Real-world problems are usually not “worst-case” but highly structured
- Fairly simple (but clever) methods work amazingly well amazingly often (though we don't really understand why)

So... Is There a Smarter Way Than Brute-Force to Solve SAT?

In theory, probably no...

- COLOURING, CLIQUE, SAT, and 1000s other problems are “all the same” — efficient algorithm for one can solve all (the problems are all **NP-complete**)
- Widely believed impossible to construct algorithms that are always (a) efficient and (b) correct (even in **worst case**)
- Settling this question is one of **Millennium Prize Problems**: Are there efficient algorithms for NP-complete problems?

In practice, definitely yes!

- Real-world problems are usually not “worst-case” but highly structured
- Fairly simple (but clever) methods work amazingly well amazingly often (though we don't really understand why)

Stark disconnect between theory and practice...

Research Goals in the MIAO Group (1/2)

Strengthen the mathematical analysis of algorithmic methods

- Study methods of reasoning powerful enough to capture state-of-the-art algorithms used in practice
- Prove theorems about their power and limitations
- E.g., resolution proof system captures CDCL reasoning

Research Goals in the MIAO Group (1/2)

Strengthen the mathematical analysis of algorithmic methods

- Study methods of reasoning powerful enough to capture state-of-the-art algorithms used in practice
- Prove theorems about their power and limitations
- E.g., resolution proof system captures CDCL reasoning

Construct stronger algorithms for combinatorial problems

- Use insights into stronger mathematical methods of reasoning to build algorithms for SAT and other combinatorial problems
- Aiming for exponential speed-ups over state of the art
- E.g., use cutting planes to build pseudo-Boolean solvers

Research Goals in the MIAO Group (2/2)

Improve understanding of efficient computation in practice

- Use computational complexity theory to study “real-world” (not worst-case) problems
- Combine theoretical study and empirical experiments
- E.g., take “crafted formulas” with provable theoretical properties and investigate correlation with practical solver performance

Research Goals in the MIAO Group (2/2)

Improve understanding of efficient computation in practice

- Use computational complexity theory to study “real-world” (not worst-case) problems
- Combine theoretical study and empirical experiments
- E.g., take “crafted formulas” with provable theoretical properties and investigate correlation with practical solver performance

Certify correctness for modern combinatorial solvers

- In many combinatorial optimization paradigms, state-of-the-art solvers are known to be buggy
- Develop methods to make solvers output not just answer but machine-verifiable proof of correctness of this answer

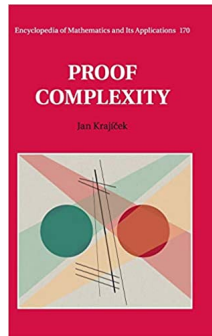
Some References for Further Reading

Handbook of Satisfiability (Especially chapter 7 ☺)



[BHvMW21]

Proof Complexity by Jan Krajíček



[Kra19]

And survey papers, slides, and videos at www.jakobnordstrom.se

Take-Home Message

- Modern SAT solvers, although based on old and simple DPLL method, can be enormously efficient in practice
- SAT solving more of an art form than a science — theoretical understanding lagging far behind
- Can use proof complexity to analyze potential and limitations of SAT solvers
- And to get inspirations for algorithms based on stronger methods of reasoning
- Lots of challenging work for PhD students and postdocs (we're hiring!)

Take-Home Message

- Modern SAT solvers, although based on old and simple DPLL method, can be enormously efficient in practice
- SAT solving more of an art form than a science — theoretical understanding lagging far behind
- Can use proof complexity to analyze potential and limitations of SAT solvers
- And to get inspirations for algorithms based on stronger methods of reasoning
- Lots of challenging work for PhD students and postdocs (we're hiring!)

Thanks for listening!

References I

- [ABdR⁺21] Albert Atserias, Ilario Bonacina, Susanna F. de Rezende, Massimo Lauria, Jakob Nordström, and Alexander Razborov. Clique is hard on average for regular resolution. *Journal of the ACM*, 68(4):23:1–23:26, August 2021. Preliminary version in *STOC '18*.
- [AS09] Gilles Audemard and Laurent Simon. Predicting learnt clauses quality in modern SAT solvers. In *Proceedings of the 21st International Joint Conference on Artificial Intelligence (IJCAI '09)*, pages 399–404, July 2009.
- [AS12] Gilles Audemard and Laurent Simon. Refining restarts strategies for SAT and UNSAT. In *Proceedings of the 18th International Conference on Principles and Practice of Constraint Programming (CP '12)*, volume 7514 of *Lecture Notes in Computer Science*, pages 118–126. Springer, October 2012.
- [BCMM05] Paul Beame, Joseph C. Culberson, David G. Mitchell, and Cristopher Moore. The resolution complexity of random graph k -colorability. *Discrete Applied Mathematics*, 153(1-3):25–47, December 2005.
- [BHvMW21] Armin Biere, Marijn J. H. Heule, Hans van Maaren, and Toby Walsh, editors. *Handbook of Satisfiability*, volume 336 of *Frontiers in Artificial Intelligence and Applications*. IOS Press, 2nd edition, February 2021.

References II

- [BIS07] Paul Beame, Russell Impagliazzo, and Ashish Sabharwal. The resolution complexity of independent sets and vertex covers in random graphs. *Computational Complexity*, 16(3):245–297, October 2007. Preliminary version in *CCC '01*.
- [Bla37] Archie Blake. *Canonical Expressions in Boolean Algebra*. PhD thesis, University of Chicago, 1937.
- [BLLM14] Armin Biere, Daniel Le Berre, Emmanuel Lonca, and Norbert Manthey. Detecting cardinality constraints in CNF. In *Proceedings of the 17th International Conference on Theory and Applications of Satisfiability Testing (SAT '14)*, volume 8561 of *Lecture Notes in Computer Science*, pages 285–301. Springer, July 2014.
- [BN21] Samuel R. Buss and Jakob Nordström. Proof complexity and SAT solving. In Biere et al. [BHvMW21], chapter 7, pages 233–350.
- [BS97] Roberto J. Bayardo Jr. and Robert Schrag. Using CSP look-back techniques to solve real-world SAT instances. In *Proceedings of the 14th National Conference on Artificial Intelligence (AAAI '97)*, pages 203–208, July 1997.
- [CCT87] William Cook, Collette Rene Coullard, and György Turán. On the complexity of cutting-plane proofs. *Discrete Applied Mathematics*, 18(1):25–38, November 1987.

References III

- [Chv73] Vašek Chvátal. Edmonds polytopes and a hierarchy of combinatorial problems. *Discrete Mathematics*, 4(1):305–337, 1973.
- [CK05] Donald Chai and Andreas Kuehlmann. A fast pseudo-Boolean constraint solver. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 24(3):305–317, March 2005. Preliminary version in *DAC '03*.
- [Coo71] Stephen A. Cook. The complexity of theorem-proving procedures. In *Proceedings of the 3rd Annual ACM Symposium on Theory of Computing (STOC '71)*, pages 151–158, May 1971.
- [CS88] Vašek Chvátal and Endre Szemerédi. Many hard examples for resolution. *Journal of the ACM*, 35(4):759–768, October 1988.
- [DLL62] Martin Davis, George Logemann, and Donald Loveland. A machine program for theorem proving. *Communications of the ACM*, 5(7):394–397, July 1962.
- [DP60] Martin Davis and Hilary Putnam. A computing procedure for quantification theory. *Journal of the ACM*, 7(3):201–215, 1960.

References IV

- [EGG⁺18] Jan Elffers, Jesús Giráldez-Cru, Stephan Gocht, Jakob Nordström, and Laurent Simon. Seeking practical CDCL insights from theoretical SAT benchmarks. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI '18)*, pages 1300–1308, July 2018.
- [EN18] Jan Elffers and Jakob Nordström. Divide and conquer: Towards faster pseudo-Boolean solving. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI '18)*, pages 1291–1299, July 2018.
- [EN20] Jan Elffers and Jakob Nordström. A cardinal improvement to pseudo-Boolean solving. In *Proceedings of the 34th AAAI Conference on Artificial Intelligence (AAAI '20)*, pages 1495–1503, February 2020.
- [Gom63] Ralph E. Gomory. An algorithm for integer solutions of linear programs. In R.L. Graves and P. Wolfe, editors, *Recent Advances in Mathematical Programming*, pages 269–302. McGraw-Hill, New York, 1963.
- [Hak85] Armin Haken. The intractability of resolution. *Theoretical Computer Science*, 39(2-3):297–308, August 1985.

References V

- [KN20] Janne I. Kokkala and Jakob Nordström. Using resolution proofs to analyse CDCL solvers. In *Proceedings of the 26th International Conference on Principles and Practice of Constraint Programming (CP '20)*, volume 12333 of *Lecture Notes in Computer Science*, pages 427–444. Springer, September 2020.
- [Kra19] Jan Krajíček. *Proof Complexity*, volume 170 of *Encyclopedia of Mathematics and Its Applications*. Cambridge University Press, March 2019.
- [Lev73] Leonid A. Levin. Universal sequential search problems. *Problemy peredachi informatsii*, 9(3):115–116, 1973. In Russian. Available at <http://mi.mathnet.ru/ppi914>.
- [LP10] Daniel Le Berre and Anne Parrain. The Sat4j library, release 2.2. *Journal on Satisfiability, Boolean Modeling and Computation*, 7:59–64, July 2010.
- [Mil00] The Millennium Problems of the Clay Mathematics Institute, May 2000. See <https://www.claymath.org/millennium-problems>.
- [MMZ⁺01] Matthew W. Moskewicz, Conor F. Madigan, Ying Zhao, Lintao Zhang, and Sharad Malik. Chaff: Engineering an efficient SAT solver. In *Proceedings of the 38th Design Automation Conference (DAC '01)*, pages 530–535, June 2001.

References VI

- [MN14] Mladen Mikša and Jakob Nordström. Long proofs of (seemingly) simple formulas. In *Proceedings of the 17th International Conference on Theory and Applications of Satisfiability Testing (SAT '14)*, volume 8561 of *Lecture Notes in Computer Science*, pages 121–137. Springer, July 2014.
- [MS99] João P. Marques-Silva and Karem A. Sakallah. GRASP: A search algorithm for propositional satisfiability. *IEEE Transactions on Computers*, 48(5):506–521, May 1999. Preliminary version in *ICCAD '96*.
- [PD07] Knot Pipatsrisawat and Adnan Darwiche. A lightweight component caching scheme for satisfiability solvers. In *Proceedings of the 10th International Conference on Theory and Applications of Satisfiability Testing (SAT '07)*, volume 4501 of *Lecture Notes in Computer Science*, pages 294–299. Springer, May 2007.
- [Rob65] John Alan Robinson. A machine-oriented logic based on the resolution principle. *Journal of the ACM*, 12(1):23–41, January 1965.
- [Spe10] Ivor Spence. sgen1: A generator of small but difficult satisfiability benchmarks. *Journal of Experimental Algorithmics*, 15:1.2:1–1.2:15, March 2010.

References VII

- [SS77] Richard M. Stallman and Gerald J. Sussman. Forward reasoning and dependency-directed backtracking in a system for computer-aided circuit analysis. *Artificial Intelligence*, 9(2):135–196, October 1977.
- [Urq87] Alasdair Urquhart. Hard examples for resolution. *Journal of the ACM*, 34(1):209–219, January 1987.
- [VS10] Allen Van Gelder and Ivor Spence. Zero-one designs produce small hard SAT instances. In *Proceedings of the 13th International Conference on Theory and Applications of Satisfiability Testing (SAT '10)*, volume 6175 of *Lecture Notes in Computer Science*, pages 388–397. Springer, July 2010.